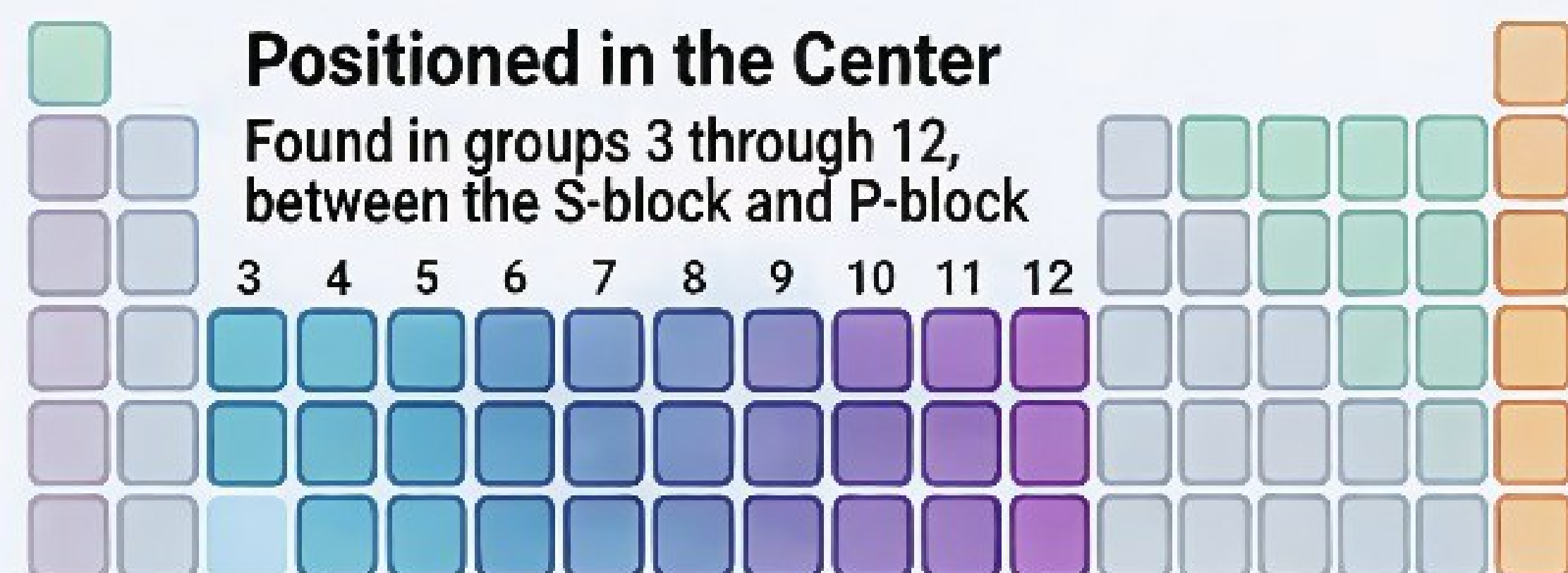


The Ultimate Guide to D & F Block Elements

THE WORLD OF D-BLOCK ELEMENTS



Why "Transition" Elements?

Form a bridge between highly electropositive S block metals and less electropositive P-block elements.



The Official Definition

An element with an incompletely filled d-orbital in its atomic state or any common oxidation state.



The Exception: Zinc (Zn), Cadmium (Cd) & Mercury (Hg)

Not considered transition elements; have completely filled d-orbital (d^{10}) in both atomic and common +2 oxidation states.

THE SUPER TRICK: IT'S ALL ABOUT UNPAIRED ELECTRONS

The More Unpaired Electrons, the Stronger the Properties

Unpaired d-orbital electrons are key to many properties.

H					
Li	Be				
Na	Mg				
K	Ca	Sc	Ti		
Rb	Sr	Y	Zr		
Cs	Ba		Hf		
Fr	Ra		Rf		

Chain of Effects from Unpaired Electrons



Strong Metallic Bonding

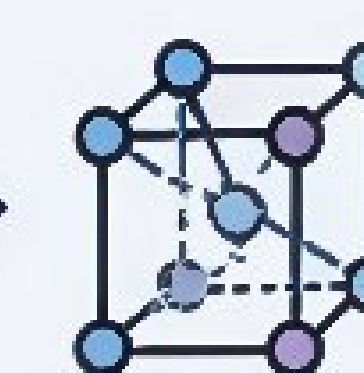
La	Ce	Pr	Nd	Pm	Eu	Gd	Th	Ho	Er	Tm	Yb	Lu
Ac	Th	Pa	U	Np	Pm	Cm	Sk	Es	Fm	Md	Nu	Lr



Hardness:
Metals are very hard and tough.



High Melting Points:
Requires a lot of energy to break bonds.



High Enthalpy of Atomization:
Difficult to break into individual atoms.



Paramagnetism:
Attracted by a magnetic field.



Colored Compounds:
Variety of colors due to d-d electron transitions.



The Opposite: Zero Unpaired Electrons

Full d^{10} configuration leads to weak metallic bonding, softer metals, lower melting points (e.g., Mercury is liquid).



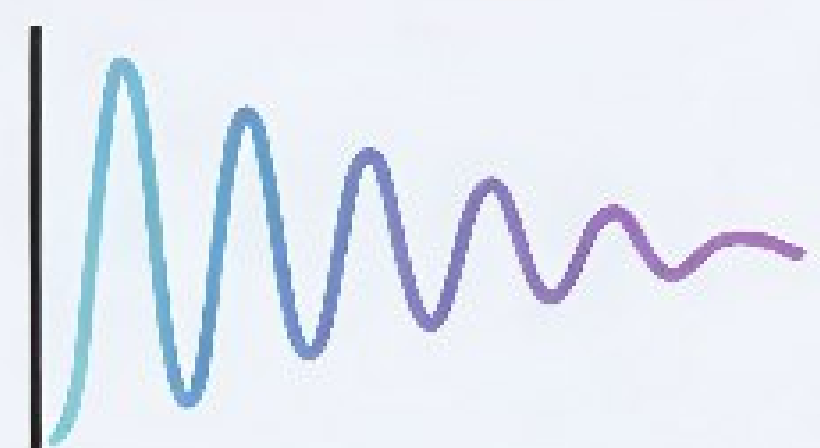
The Manganese (Mn) Anomaly

Despite d^5 (many unpaired electrons), has unusually low melting point: stable, stable, half-filled configuration makes it reluctant to form strong metallic bonds.



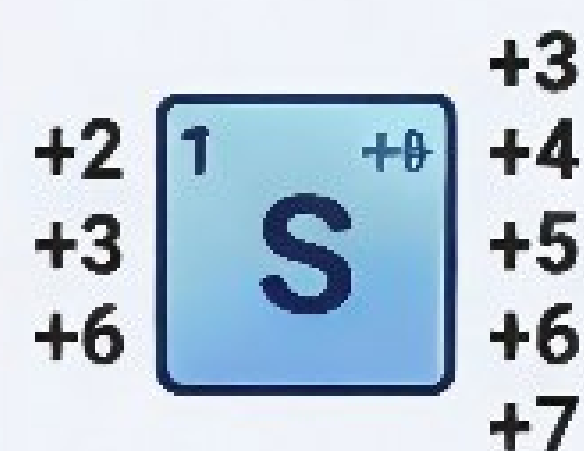
KEY TRENDS AND PROPERTIES

Atomic Radii Trend Across the Series



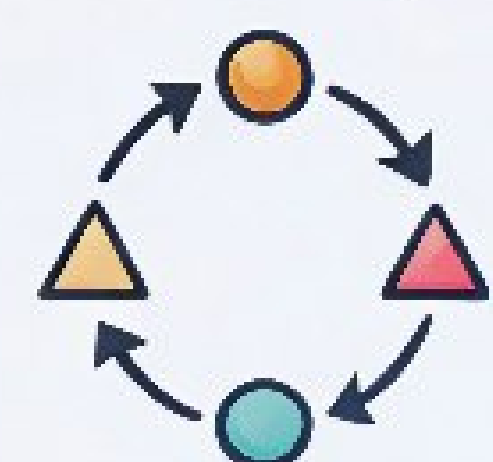
Decreases (increased nuclear charge), then almost constant (electron repulsion balances charge), finally increases slightly (repulsion dominates).

Variable Oxidation States



Caused by small energy difference between ns and $(n-1)d$ orbitals, allowing electrons from both to participate in bonding (Mn shows +2 to +7).

Why are They Great Catalysts?



Variable oxidation states allow participation in diverse chemical reactions (e.g., Iron, V_2O_5).

Ionization Enthalpy: The Cost of Removing an Electron



Energy required to remove an electron. High for stable half-filled (d^5) or fully-filled (d^{10}) configurations.

A QUICK LOOK AT THE F-BLOCK

Two Series: Lanthanoids and Actinoids

Lanthanoids: 14 elements following Lanthanum (La).
Actinoids: 14 elements following Actinium (Ac).

Lanthanoid Contraction: A Crucial Concept



Cause: Poor shielding effect of 4f-electrons; nucleus pulls outer electrons more strongly, causing the atom to shrink.

Lanthanoid Contraction

Elements of 4d and 5d series have nearly identical atomic radii and properties (e.g., Zirconium and Hafnium are almost chemical twins).

Consequence of Lanthanoid Contraction



Elements of 4d and 5d series have nearly identical atomic radii and properties (e.g., Zr and Hf are almost chemical twins).

Actinoid Contraction

Similar size decrease in Actinoid series due to poor shielding effect of 5f-electrons.